54.1: Interactions between species can help, harm, or have no effect on the individuals involved

Some key relationships in the life of an organism are its interactions with individuals of other species in the community. These **interspecific interactions** include competition, predation, herbivory, parasitism, mutualism, and commensalism. In this section, we'll define and describe each of these interactions, grouping them according to whether they have positive (+) or negative (–) effects on the survival and reproduction of individuals engaged in the interaction.

For example, predation is a +/– interaction, with a positive effect on the survival and reproduction of members of the predator population and a negative effect on members of the prey population. Mutualism is a +/+ interaction in which the survival and reproduction of individuals of each species is increased in the presence of the other. A zero (0) indicates that the members of a species are not affected by the interaction.

We'll consider three broad categories of ecological interactions: competition (–/–), exploitation (+/–), and positive interactions (+/+ or +/0). As you study examples of these interactions, bear in mind that their effects can change over time. For example, an interaction that typically benefits individuals of both species may at times continue to benefit one of them, while harming or having no effect on the other.

Competition

Competition is a –/– interaction that occurs when individuals of different species each use a resource that limits the survival and reproduction of both individuals. (As described in Figure 53.18, competition for resources can also occur between members of the same species; this case is sometimes distinguished as *intraspecific competition*.) Weeds growing in a garden compete with garden plants for soil nutrients and water. Lynx and foxes in the northern forests of Alaska and Canada compete for prey such as snowshoe hares. In contrast, some resources, such as oxygen, are rarely in short supply on land; most terrestrial species use this resource but do not usually compete for it.

Competitive Exclusion

What happens in a community when individuals of two species compete for limited resources? In 1934, Russian ecologist G. F. Gause studied this question using laboratory experiments with two closely related ciliate species, *Paramecium aurelia* and *Paramecium caudatum* (see Figure 28.19a). He cultured the

species under stable conditions, adding a constant amount of food each day. When Gause grew the two species separately, each population increased rapidly in number and then leveled off at the apparent carrying capacity of the culture (see Figure 53.10a for an illustration of the logistic growth of a *Paramecium* population). But when Gause grew the two species together, *P. caudatum* became extinct in the culture. Gause inferred that *P. aurelia* had a competitive edge in obtaining food. More generally, his results led him to conclude that two species whose members compete for the same limiting resources cannot coexist permanently in the same place. In the absence of disturbance, one species will use the resources more efficiently and reproduce more rapidly than the other. Even a slight reproductive advantage will eventually lead to local elimination of the inferior competitor, an outcome called **competitive exclusion**.

Ecological Niches and Natural Selection

EVOLUTION Competition for limited resources can cause evolutionary change in populations. One way to examine how this occurs is to focus on an organism's **ecological niche**, the specific set of biotic and abiotic resources that an organism uses in its environment. The niche of a tropical tree lizard, for instance, includes the temperature range it tolerates, the size of branches on which it perches, the time of day when it is active, and the sizes and kinds of insects it eats. Such factors define the lizard's niche, or ecological role—how it fits into an ecosystem.

We can use the niche concept to restate the principle of competitive exclusion: Two species cannot coexist permanently in a community if their niches are identical. However, ecologically similar species *can* coexist in a community if one or more significant differences in their niches arise through time. Evolution by natural selection can result in one of the species using a different set of resources or similar resources at different times of the day or year. The differentiation of niches that enables similar species to coexist in a community is called **resource partitioning(Figure 54.2)**.

Figure 54.2

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Resource partitioning among Dominican Republic lizards.
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Forest labeled with different use of resources by seven species of Anolis lizards.

Seven species of Anolis lizards live in close proximity, and all feed on insects and other small arthropods. However, competition for food is reduced because each lizard species has a different preferred perch, thus occupying a distinct niche.

Forest labeled with different use of resources by seven species of Anolis lizards. A distichus perches on fence posts and other sunny surfaces. A insolitus usually perches on shady branches. A ricordi is shown high in the

tree canopy. A christophei is shown at the mid-level of the tree trunks. A cybotes is shown at the lower tree trunk level. A etheridgei is just above the level of the forest floor. A aliniger is shown just above the level of the fence.

As a result of competition, a species' *fundamental niche*, which is the niche potentially occupied by that species, is often different from its *realized niche*, the portion of its fundamental niche that it actually occupies. Ecologists can identify the fundamental niche of a species by testing the range of conditions in which it grows and reproduces in the absence of competitors. They can also test whether a potential competitor limits a species' realized niche by removing the competitor and seeing if the first species expands into the newly available space. The classic experiment depicted in **Figure 54.3** clearly showed that competition between individuals of two barnacle species kept one species from occupying part of its fundamental niche.

Figure 54.3

Inquiry: Can a species' niche be influenced by competition?

Experiment

Ecologist Joseph Connell studied two barnacle species—*Chthamalus stellatus* and *Balanus balanoides*—that have a stratified distribution on rocks along the coast of Scotland. *Chthamalus* is usually found higher on the rocks than *Balanus*. To determine whether the distribution of *Chthamalus* is the result of competition with *Balanus*, Connell removed *Balanus* from the rocks at several sites.

A diagram of competition between barnacle species.

Experiment: Chthamalus is distributed higher on the rocks than Balanus, with realized niche of Chthamalus near high tide line. Realized niche of Balanus is near low tide line.

Results

Chthamalus spread into the region formerly occupied by *Balanus*.

Results of an experiment on competition between barnacle species.

Results: With *Balanus* absent, *Chthamalus* fundamental niche spreads down from high tide to the low tide line.

Conclusion

Competition makes the realized niche of *Chthamalus* much smaller than its fundamental niche.

Data from J. H. Connell, The influence of interspecific competition and other factors on the distribution of the barnacle *Chthamalus stellatus, Ecology* 42:710–723 (1961).

WHAT IF?_Other observations showed that Balanus cannot survive high on the rocks because it dries out during low tides. How would Balanus's realized niche compare with its fundamental niche?_

For suggested answer, see*Appendix A*.

Instructors: A related Experimental Inquiry Tutorial can be assigned in **Mastering Biology**.

Species can partition their niches not just in space, as lizards and barnacles do, but in time as well. The common spiny mouse (*Acomys cahirinus*) and the golden spiny mouse (*A. russatus*; **Figure 54.4**) live in rocky habitats of the Middle East and Africa, using similar microhabitats and food sources. Where they coexist, *A. cahirinus* is nocturnal (active at night), while *A. russatus* is diurnal (active during the day). Surprisingly, laboratory research showed that *A. russatus* is naturally nocturnal. To be active during the day, it must override its biological clock in the presence of *A. cahirinus*. When researchers in Israel removed all *A. cahirinus* individuals from a site in the species' natural habitat, *A. russatus* individuals at that site became nocturnal, consistent with the laboratory results. This change in behavior suggests that these individuals were competing for resources and that partitioning of their active time helps them coexist.

Figure 54.4

The golden spiny mouse (*Acomys russatus*).

A golden spiny mouse.

Character Displacement

Closely related species whose populations are sometimes allopatric (geographically separate; see Concept 24.2) and sometimes sympatric (geographically overlapping) provide additional evidence of how competition affects communities. In some cases, the allopatric populations of such species are morphologically similar and use similar resources. By contrast, sympatric populations, which would potentially compete for resources, show differences in body structures and in the resources they use. This tendency for characteristics to diverge more in sympatric than in allopatric populations of two species is called **character displacement**. An example of character displacement can be seen in two species of Galápagos finches, *Geospiza fuliginosa* and *Geospiza fortis*: Beak depths in these species are similar in allopatric populations but have diverged considerably in sympatric populations **(Figure 54.5)**.

Figure 54.5

Character displacement: indirect evidence of past competition.

Graph of percent of individuals in each size class versus beak depth for G. fulginosa, allopatric on Los Hermanos, G. fortis, allopatric on Daphne and sympatric populations on Santa María and San Cristóbal.

Allopatric populations of Geospiza fuliginosa and Geospiza fortis on Los Hermanos and Daphne Islands have similar beak morphologies (top two graphs) and presumably eat similarly sized seeds. However, where the two species are sympatric on Floreana and San Cristóbal, G. fuliginosa has a shallower, smaller beak and G. fortis a deeper, larger one (bottom graph), adaptations that favor eating different-sized seeds.

Graph of percent of individuals in each size class versus beak depth. Beak depth (millimeters) is on the x-axis from 6 to 16 with increments every 1 millimeter. Beak depth is measured as the distance from the lower base of the beak to the upper base. Percent of individuals in each size class from 0 to 60 is on the y-axis. The 3 populations are G. fulginosa, allopatric on Los Hermanos, G. fortis, allopatric on Daphne and sympatric populations on Santa Maria and San Cristóbal. See Table 1 for data.

WHAT IF? *If* G. fuliginosa *became extinct on Floreana, predict how beak depth would change over time in Floreana's* G. fortis *population. Explain your reasoning.*

*For suggested answer, see*Appendix A.**

Exploitation

All nonphotosynthetic organisms must eat, and all organisms are at risk of being eaten. As a result, much of the drama in nature involves **exploitation**, a term for any type of +/− interaction in which individuals of one species benefit by feeding on (and thereby harming) individuals of the other species. Exploitative interactions include predation, herbivory, and parasitism.

Predation

Predation is a +/− interaction in which an individual of one species, the predator, kills and eats an individual of the other species, the prey. Though the term *predation* generally elicits such images as a lion attacking and eating an antelope, it applies to a wide range of interactions. A rotifer (a tiny aquatic animal that is smaller than many unicellular protists) that kills a protist by eating it can also be considered a predator. Because eating and avoiding being eaten are prerequisites to reproductive success, the adaptations of both predators and prey tend to be refined through natural selection. In the **Scientific Skills Exercise**, you can interpret data on the impact of natural selection for a specific predator-prey interaction.

Scientific Skills Exercise

Making a Bar Graph and a Scatter Plot

AP Science Practices 2, 3, 4, 6

Can a Native Predator Species Adapt Rapidly to an Introduced Prey Species? Cane toads (*Bufo marinus*) were introduced to Australia in 1935 in a failed attempt to control an insect pest. Since then, the toads have spread across northeastern Australia, with a population of over 200 million today. Cane toads have glands that produce a toxin that is poisonous to snakes and other potential predators. In this exercise, you will graph and interpret data from a two-part experiment conducted to determine whether native Australian predators have developed resistance to the cane toad toxin.

Decorative image.

How the Experiment Was Done In part 1, researchers collected 12 red-bellied black snakes (*Pseudechis porphyriacus*) from areas where cane toads had existed for 40–60 years and another 12 from areas free of cane toads. They recorded the percentage of snakes from each area that ate either a freshly killed native frog (*Limnodynastes peronii*, a species the snakes commonly eat) or a freshly killed cane toad from which the toxin gland had been removed (making the toad nonpoisonous). In part 2, researchers collected snakes from areas where cane toads had been present for 5–60 years. To assess how cane toad toxin affected the physiological activity of these snakes, they injected small amounts of the toxin into the snakes' stomachs and measured the snakes' swimming speed in a small pool.

Data from the Experiment, Part 1

A table illustrates types of prey offered and percent of snakes from each area that ate each type of prey.

Type of Prey Offered	Percent of Snakes from Each Area That Ate Each Type of Prey
Area with Cane Toads Present for 40 to 60 years	Area with No Cane Toads
Native frog	100
Cane toad	0

Data from the Experiment, Part 2

A table illustrates the number of years cane toads had been present in the area and percent reduction in snake swimming speed.

Data from B. L. Phillips and R. Shine, An invasive species induces rapid adaptive change in a native predator: cane toads and black snakes in Australia, *Proceedings of the Royal Society B* 273:1545–1550 (2006).

Number of Years Cane Toads Had Been Present in the Area	5	10	10	20	50	60	60	60	60	60
Percent Reduction in Snake Swimming Speed	52	19	30	30	5	5	9	11	12	22

INTERPRET THE DATA

- Make a bar graph of the data in part 1. (For additional information about graphs, see the Scientific Skills Review in Appendix D.)
- What do the data represented in the graph suggest about the effects of cane toads on the predatory behavior of red-bellied black snakes in areas where the toads are and are not currently found?
- Suppose a novel enzyme that deactivates the cane toad toxin evolved in a snake population exposed to cane toads. If the researchers repeated part 1 of this study, predict how the results would change.
- Identify the dependent and independent variables in part 2 and make a scatter plot. What conclusion would you draw about whether exposure to cane toads is having a selective effect on the snakes? Explain.
- Explain why a bar graph is appropriate for presenting the data in part 1 and a scatter plot is appropriate for the data in part 2.

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

Many important feeding adaptations of predators are obvious and familiar. Most predators have acute senses that enable them to find and identify potential prey. Rattlesnakes and other pit vipers, for example, find their prey with a pair of heat-sensing organs located between their eyes and nostrils (see Figure 50.7b). Owls have characteristically large eyes that help them see prey at night. Many predators also have adaptations such as claws, fangs, or poison that help them catch and subdue their food. Predators that pursue their prey are generally fast and agile, whereas those that lie in ambush are often disguised in their environments.

Just as predators possess adaptations for capturing prey, potential prey have adaptations that help them avoid being eaten. In animals, these adaptations include behavioral defenses such as hiding, fleeing, and forming herds or schools. Active self-defense is less common, though some large grazing mammals vigorously defend their young from predators.

Animals also display a variety of morphological and physiological defensive adaptations. Mechanical or chemical defenses protect species such as porcupines and skunks **(Figure 54.6a and b)**. Some animals, such as the European fire salamander, can synthesize toxins; others accumulate toxins passively from the plants they eat. Animals with effective chemical defenses often exhibit bright **aposematic coloration**, or warning coloration, such as that of poison dart frogs **(Figure 54.6c)**. Such coloration seems to be adaptive because predators often avoid brightly colored prey. **Cryptic coloration**, or camouflage, makes prey difficult to see **(Figure 54.6d)**.

Figure 54.6

Examples of defensive adaptations in animals.

Six types of defense.

Parts are as follows.

a. Mechanical defense. Porcupine with quills.

b. Chemical defense. Skunk.

c. Aposematic coloration: warning coloration. Poison dart frog with bright orange and blue colors.

d. Cryptic coloration: camouflage. A canyon tree frog blends in with his surroundings.

e. Batesian mimicry: A harmless species mimics a harmful one. A nonvenomous hawkmoth larva resembles a venomous green parrot snake with similar eye placement and head shape.

f. Mullerian mimicry: Two unpalatable species mimic each other. A yellow jacket and a Cuckoo bee have similar coloration patterns.

MAKE CONNECTIONS *Explain how natural selection could increase the resemblance of a harmless species to a distantly related harmful species. In addition to selection, what else could account for a harmless species resembling a closely related harmful species? (See*Concept 22.2*.)**

*For suggested answer, see*Appendix A*.**

HHMI Biointeractive Video

Moth Mimicry: Using Ultrasound to Avoid Bats

image

Play

image

Play

For more resources from HHMI BioInteractive, go to www.biointeractive.org.

Some prey species are protected by their resemblance to other species. For example, in **Batesian mimicry**, a palatable or harmless species mimics an unpalatable or harmful species to which it is not closely related. The larva of the hawkmoth *Hemeroplanes ornatus* puffs up its head and thorax when disturbed, looking like the head of a small venomous snake **(Figure 54.6e)**. In this case, the mimicry even involves behavior; the larva weaves its head back and forth and hisses like a snake. Such cases of Batesian mimicry are thought to result from natural selection, as individuals in the harmless species that happen to more closely resemble the harmful one are avoided by predators who have learned not to eat the harmful ones. Over time, closer and closer resemblance to the harmful species evolves. In **Müllerian mimicry**, two or more unpalatable species, such as the cuckoo bee and yellow jacket, resemble each other **(Figure 54.6f)**. Presumably, the more unpalatable prey there are, the faster predators learn to avoid prey with that particular appearance. Mimicry has also evolved in many predators. The mimic octopus *Thaumoctopus mimicus***(Figure 54.7)** can take on the appearance and movement of more than a dozen marine animals, including crabs, sea stars, sea snakes, fish, and stingrays. This octopus's ability to mimic other animals enables it to approach prey—for example, imitating a crab to approach another crab and eat it. The octopus also can defend itself from predators through mimicry. When attacked by a damselfish, the octopus quickly mimics a banded sea snake, a known predator of the damselfish.

Figure 54.7

The mimic octopus.

Three photos show an octopus mimicking, a, a sea snake, b, a flounder, and c, a stingray.

(a) *After hiding six of its tentacles in a hole in the seafloor, the octopus waves its other two tentacles to mimic a sea snake.* ***(b)*** *Flattening its body and arranging its arms to trail behind, the octopus mimics a flounder (a flat fish).* ***(c)*** *It can mimic a stingray by flattening most of its tentacles alongside its body while allowing one tentacle to extend behind it.*

Interview

Interview with Tracy Langkilde: Studying the evolution of novel defensive adaptations in lizards

Tracy Langkilde.

Herbivory

Herbivory is an exploitative (+/−) interaction in which an organism—an herbivore—eats parts of a plant or alga, thereby harming it but usually not killing it. While large mammalian herbivores such as cattle, sheep, and water buffalo may be most familiar, most herbivores are actually invertebrates, such as grasshoppers, caterpillars, and beetles. In the ocean, herbivores include sea urchins, some tropical fishes, and certain mammals **(Figure 54.8)**.

Figure 54.8

An herbivorous marine mammal.

A manatee.

This West Indian manatee (Trichechus manatus) in Florida is grazing on Hydrilla, an introduced plant species.

Like predators, herbivores have many specialized adaptations. Many herbivorous insects have chemical sensors on their feet that enable them to distinguish between plants based on their toxicity or nutritional value. Some mammalian herbivores, such as goats, use their sense of smell to examine plants, rejecting some and eating others. They may also eat just a specific part of a plant, such as the flowers. Many herbivores also have specialized teeth or digestive systems adapted for processing vegetation (see Concept 41.4).

Unlike animals, plants cannot run away to avoid being eaten. Instead, a plant's arsenal against herbivores may feature chemical toxins or structures such as spines and thorns. Among the plant compounds that serve as chemical defenses are the poison strychnine, produced by the tropical vine *Strychnos toxifera*; nicotine, from the tobacco plant; and tannins, from a variety of plant species. Compounds that are not toxic to humans but may be distasteful to many herbivores are responsible for the familiar flavors of cinnamon, cloves, and peppermint. Certain plants produce chemicals that cause abnormal development in some insects that eat them. For more examples of how plants defend themselves, see Make Connections Figure 39.27, "Levels of Plant Defenses Against Herbivores."

Parasitism

****Parasitism** is a +/− exploitative interaction in which one organism, the **parasite**, derives its nourishment from another organism, its **host**, which is harmed in the process. Parasites that live within the

body of their host, such as tapeworms, are called **endoparasites**; parasites that feed on the external surface of a host, such as ticks and lice, are called **ectoparasites**. Some ecologists have estimated that at least one-third of all species on Earth are parasites. In one particular type of parasitism, parasitoid insects—usually small wasps—lay eggs on or in living hosts. The larvae then feed on the body of the host, eventually killing it.

Many parasites have complex life cycles involving multiple hosts. The blood fluke, which currently infects approximately 200 million people around the world, requires two hosts at different times in its development: humans and freshwater snails (see Figure 33.10). Some parasites change the behavior of their current host in ways that increase the likelihood that the parasite will reach its next host. For instance, crustaceans that are parasitized by acanthocephalan (spiny-headed) worms leave protective cover and move into the open, where they are more likely to be eaten by the birds that are the second host in the worm's life cycle.

Parasites can significantly affect the survival, reproduction, and density of their host population, either directly or indirectly. For example, ticks that feed as ectoparasites on moose can weaken their hosts by withdrawing blood and causing hair breakage and loss. In their weakened condition, the moose have a greater chance of dying from cold stress or predation by wolves.

Positive Interactions

While nature abounds with dramatic and gory examples of exploitative interactions, ecological communities are also heavily influenced by **positive interactions**, a term that refers to a +/+ or +/0 interaction between members of two species in which at least one individual benefits and neither is harmed. Positive interactions include mutualism and commensalism. As we'll see, positive interactions can affect the diversity of species found in an ecological community.

Mutualism

Mutualism is a +/+ interaction that benefits individuals of both of the interacting species. Mutualisms are common in nature, as illustrated by examples seen in previous chapters, including cellulose digestion by microorganisms in the digestive systems of termites and ruminant mammals, animals that pollinate flowers or disperse seeds, nutrient exchange between fungi and plant roots in mycorrhizae, and photosynthesis by unicellular algae in corals. In some mutualisms, such as the acacia-ant example shown in **Figure 54.9**, each of the interacting individuals depends on the other for their survival and reproduction. In other mutualisms, however, both individuals can survive on their own.

Figure 54.9

Mutualism between acacia trees and ants.

Part a. Acacia with protein-rich swelling. Part b. Acacia growing with vegetation clipped from area surrounding tree.

Part a. Certain species of acacia trees in Central and South America have hollow thorns (not shown) that house stinging ants of the genus Pseudomyrmex. The ants feed on nectar produced by the tree and on protein-rich swellings (yellow in the photograph) at the tips of leaflets.

Part b. The acacia benefits because the pugnacious ants, which attack anything that touches the tree, remove fungal spores, small herbivores, and debris. They also clip vegetation that grows close to the acacia.

Typically, both partners in a mutualism incur costs as well as benefits. In mycorrhizae, for example, the plant often transfers carbohydrates to the fungus, while the fungus transfers limiting nutrients, such as phosphorus, to the plant. Each partner benefits, but each partner also experiences a cost: It transfers materials that it could have used to support its own growth and metabolism. For an interaction to be considered a mutualism, the benefits to each partner must exceed the costs. When this is not the case, the mutualism may break down, at least temporarily. In some mycorrhizae, for example, the plant may cease to supply carbohydrates to its fungal partner when soil nutrients are plentiful—a change that occurs because the cost of supporting the fungus has become greater than the benefits the fungus can provide.

Commensalism

An interaction that benefits the individuals of one of the interacting species but neither harms nor helps the individuals of the other species is called **commensalism** (+/0). Like mutualism, commensalisms are common in nature. For instance, many wildflowers that grow best in low light levels are found only in shaded, forest floor environments. Such shade-tolerant “specialists” depend entirely on the trees that tower above them—the trees provide their dim habitat. Yet the survival and reproduction of the trees are not affected by these wildflowers. Thus, these species are involved in a +/0 interaction in which the wildflowers benefit and the trees are not affected.

In another example of a commensalism, cattle egrets feed on insects flushed out of the grass by grazing bison, cattle, horses, and other herbivores **(Figure 54.10)**. Because the birds increase their feeding rates when following the herbivores, they clearly benefit from the association. Much of the time, the herbivores are not affected by the birds. At times, however, the herbivores too may derive some benefit; for example, the birds may remove and eat ticks and other ectoparasites from the herbivores’ skin, or they may warn the herbivores of a predator’s approach. This example provides another illustration of a key point about ecological interactions: Their effects can change over time. In this case, an interaction whose effects are typically +/0 (commensalism) may at times become +/+ (mutualism).

Figure 54.10

Commensalism between cattle egrets and African buffalo.

Cattle egrets perch on and near an African buffalo in the dry grass.

Positive interactions can have major effects on ecological communities. For instance, the black rush *Juncus gerardii* alters soil conditions in ways that benefit other plant species in New England salt marshes **(Figure 54.11a)**. *Juncus* shades the soil surface, which reduces evaporation and thus lowers the concentration of salt in the soil. The presence of *Juncus* also increases soil oxygen levels; this occurs because some oxygen leaks into the soil as *Juncus* transports oxygen to its belowground tissues. In one study, when *Juncus* was removed from areas of the marsh, those areas supported 50% fewer plant species **(Figure 54.11b)**.

Figure 54.11

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Positive interactions in New England salt marshes.
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Part a. Salt marsh with Juncus in foreground. Part b. Bar chart showing number of plant species with and without Juncus present. With Juncus, 8 plant species are recorded. Without Juncus, 4 are recorded.

When black rush (Juncus gerardii) is present, soil salt concentrations drop and soil oxygen levels rise, increasing the number of plant species that can live in the marsh.

Like positive interactions, competition and exploitation (predation, herbivory, and parasitism) also can strongly affect ecological communities, as examples throughout the rest of this chapter will show.